

Umair Ul Hassan

Full-Stack Developer — Web & Mobile

+92 347 3256547 | Lahore, Pakistan | umairulhassan.cs@gmail.com | github.com/umairulhassan | linkedin.com/in/umair-ul-hassan-cs | umairulhassan.site

Full-Stack Developer building web and mobile applications across SaaS, commerce, ERP, and applied AI. Experienced with TypeScript/JavaScript, React, Next.js, Node.js, React Native, PostgreSQL, MongoDB, and Python-based AI services, with hands-on experience in API development, production UI, computer vision, AR, authentication, payments, and real-time systems.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, Java
Web / Backend	React, Next.js, Node.js, Express.js, FastAPI, REST APIs
Mobile / 3D	React Native, Expo, Three.js, Viro React
AI / ML	PyTorch, Vision Transformers (ViT), CNNs, LLM & Speech Integrations
Data / Cloud	PostgreSQL, MongoDB, Firebase Suite
Tools	Git, GitHub, Tailwind CSS

EXPERIENCE

Web Development Intern

ARCH TECHNOLOGIES

React, Next.js, Node.js, JavaScript, REST APIs

Aug 2025 – Oct 2025 | Remote, Pakistan

- Developed frontend and backend features for live web projects using React, Next.js, and Node.js based on assigned development tasks.
- Built responsive UI components and API endpoints, integrating frontend workflows with backend services and database records.
- Diagnosed and resolved production issues while addressing application reliability and performance-related problems.

SELECTED PROJECTS

GardenMate Ecosystem

FINAL YEAR PROJECT

React Native, Expo, PyTorch (ViT-B/16), Viro React AR, FastAPI, Firebase, Stripe

2025 – 2026

- Built an AI-powered, AR-integrated gardening assistant and plant marketplace with 12+ specialized modules across four role-based portals (Customer, Vendor, Rider, Admin).
- Fine-tuned a Vision Transformer (ViT-B/16) model for 69 plant species and combined it with the PlantNet API in a 60/40 weighted ensemble, achieving ~350ms inference latency.
- Implemented Viro React AR plane detection for a "Place Before You Buy" 3D plant-placement experience and a camera Snap-to-Autolist product listing workflow.
- Designed the marketplace commission engine with Stripe payment settlement and automated delivery fee calculations across Customer, Vendor, Rider, and Admin portals.

Vouch Keychain

PERSONAL SAAS PRODUCT

Next.js 16, React 19, TypeScript, Firebase Suite, Twilio, Resend, Three.js

2026

- Built and deployed a QR-based lost-item recovery SaaS product with privacy-preserving masked-contact communication, currently used by around 10 customers.
- Implemented real-time location-recovery telemetry, multi-channel alert dispatch via Twilio SMS/voice and Resend email, and an interactive Three.js 3D product visualizer.

Evershine Academy ERP & LMS

CLIENT PROJECT

Next.js 16, React 19, TypeScript, PostgreSQL

2026

- Developed ERP/LMS features alongside the team for a deployed multi-campus education system used by 100+ users.
- Built and integrated 7 role-based access portals, automated PDF fee-challan generation, staff payroll workflows, and on-premise biometric fingerprint attendance synchronization.

NutriLens AI

React Native, Expo SDK 54, Zustand 5, Firebase Realtime Database, OpenRouter AI2025

- Built a cross-platform mobile nutrition application combining food photography and grocery barcode inputs for automated macronutrient tracking.
- Implemented clinical Mifflin-St Jeor metabolic expenditure calculations (BMR/TDEE) and an interactive AI dietitian coach powered by OpenRouter AI.

FinGo Marketplace

React.js, Node.js, Express.js, MongoDB, JWT2024

- Built a multi-vendor commerce platform connecting retail and grocery stores with customers across Lahore, Karachi, and Islamabad.
- Implemented 3-tier role-based access control, secure JWT authentication with refresh-token rotation, and real-time inventory tracking.

Voice AI Copilot

Groq LLaMA 3.3 70B, Groq Whisper Large v3, Web Audio API2025 – 2026

- Built a browser-based voice assistant converting spoken commands into structured task and reminder actions through LLM tool calling.
- Integrated speech recognition and LLM processing using Groq Whisper Large v3 and LLaMA 3.3 70B with the native browser Web Audio API.

ACHIEVEMENTS

Digital Pakistan Speed Programming Competition (DPSPC)TOP 50 NATIONWIDE

National Speed Programming & Algorithmic Problem Solving Contest (~450 Teams)2025

- Ranked among the top 50 teams nationwide out of ~450 competing teams in speed programming and algorithmic problem solving under strict time constraints.

InnovateX 2.0 Coding CompetitionGIFT UNIVERSITY

Senior-Level Algorithmic Competitive Programming Hackathon2025

- Competed in senior-level algorithmic coding challenges evaluating data structures, optimal time/space complexity, and efficient algorithm design.

EDUCATION

Bachelor of Science in Computer Science (BS CS)GIFT UNIVERSITY, GUJRANWALA

Coursework: Software Engineering, Data Structures & Algorithms, Database Systems, Artificial Intelligence, Machine Learning, Computer Vision, Distributed Systems2022 – 2026